

ESP8266 Wi-Fi Microchip – Building a Wi-Fi Hacking Toolkit

Project Description

The **ESP8266** is a low-cost Wi-Fi microchip used widely in **IoT devices** such as smart home systems, energy monitors, industrial automation, and medical sensors. This project repurposes the ESP8266 (e.g., NodeMCU or Wemos D1 Mini) into a **Wi-Fi hacking toolkit** for **educational and ethical hacking** purposes.

The project demonstrates how vulnerabilities in unsecured or misconfigured Wi-Fi networks can be tested and visualized using a compact and programmable microcontroller.

Disclaimer

This toolkit is designed **strictly for educational and ethical security research** purposes only. Unauthorized access or network interference is illegal and unethical. Always have **explicit permission** before testing any network.

Components Used

- **ESP8266 (NodeMCU / Wemos D1 Mini)** – Wi-Fi microcontroller board
 - **Micro-USB Cable** – For programming and power
 - **Computer with Arduino IDE / PlatformIO** – For firmware upload
 - **Wi-Fi Deauther Firmware** – Open-source firmware for security testing
 - **Battery Pack / Power Bank** (optional) – For portable deployment
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Firmware Tools / Libraries

- **ESP8266 Deauther** – Open-source firmware that allows you to perform Wi-Fi network scans, deauthentication attacks, and beacon spam (only on test networks)
- **Arduino IDE** – For flashing and modifying the firmware
- **ESP8266 Board Support** – Installed in Arduino IDE via Board Manager
- **Libraries:** ESP8266WiFi, Adafruit_GFX, OLED SSD1306 (optional for screen display)

Working Principle

- The **ESP8266** scans nearby Wi-Fi networks and devices (clients).
- It allows the user to send **deauthentication frames** (disconnect signals) to test how clients react.
- It can also **broadcast fake access points (APs)** to simulate phishing or rogue network scenarios.
- The toolkit is controlled via a built-in web interface served by the ESP8266 itself.
- Users can access it by connecting to the ESP8266's Wi-Fi and opening a browser at 192.168.4.1.

Key Features

- Scans and lists Wi-Fi access points and connected devices
- Sends deauthentication packets (disconnect clients)
- Creates fake access points for testing phishing defenses
- Control via a simple web interface (no external apps needed)
- Entirely portable and USB-powered

Applications

- Ethical hacking and Wi-Fi penetration testing
- Network security awareness and demonstrations
- IoT network vulnerability education
- Educational workshops on wireless security
- Testing robustness of captive portals and IoT networks

Precautions and Legal Use

- **Only use on test networks** you own or are authorized to test
 - Deauthentication attacks can cause network denial of service
 - Avoid disrupting public Wi-Fi, neighbor's routers, or corporate networks
 - Use a **Faraday cage**, Wi-Fi isolator box, or RF lab for controlled demos
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Power Supply Notes

- Can run from USB power, 5V battery bank, or Li-ion cell
- Use deep sleep mode if deploying for long sessions
- Optional: Attach OLED screen for status monitoring